Project Proposal

Team Project Module · National College of Ireland · Individual Project

Project Name: PrepCast
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1. Objectives & Problem

In the food service sector, managers must organise how many meals they need to prepare for each week. While forecasting systems exist, they are too expensive for small to medium sized operations (SMEs), leading to decisions based on experience, past orders, and manual assumptions. Without accurate forecasting, food service businesses can find themselves under-prepared, missing out on orders, or over-stocked, resulting in wastage.

This project aims to build **PrepCast**, an Artificial Intelligence (AI)-powered forecasting system, utilising Machine Learning (ML) models, to help with accurately predicting how many meals will need to be prepared for the coming week. It will factor in, not only previous weekly sales, but also seasonal changes and promotions.

This system will help managers meet demand, avoid wastage, and offer an affordable AI solution to help their business.

2. Background & Related Work

While manual forecasting is a simplified approach, it does have a certain level of success with routine weeks. Where it struggles is when new factors are added like holidays or promotions. There are numerous enterprise solutions on the market such as SAP and Oracle that are powerful, but these systems come with a high cost, especially to an SME operation.

Within forecasting methods, statistical models like ARIMA score well in accuracy but they can struggle with promotions and other special factors [1]. A recent retail forecasting competition found that tree-based ML models outperformed ARIMA models on prediction accuracy [2].

PrepCast will bring ML prediction techniques as a more affordable and accurate forecasting solution for managers to use in their planning.

3. Proposed Solution

PrepCast is a machine learning tool to give recommendations to food-service managers of how many meals to prepare for the upcoming week.

Core Features

- Weekly demand forecast, per centre: ML model trained on past orders, price, promotions and seasonality
- Preparation Recommendation: Turns the forecast into quantity to prepare for the upcoming week.
- Adjust and save plan: the manager reviews the recommendation, tweaks if needed, and saves the plan
- Records actual orders: Manager saves the sales back to the database where the model can retrain on current data.

Minimum Viable Product (MVP) – Week 8

A trained ML model that displays its prediction of the upcoming week's preparation list, per centre. The adjust-plan and record-actuals features follow for the final submission.

4. Technical Approach

PrepCast is a web app using HTML/Tailwind/JavaScript for frontend and Flask/Python/REST API, with data stored in an SQLite database.

An ML model will be used for the forecasting system. Several regression models, e.g. Random Forest, Multiple Linear Regression, will be tested, tuned, and evaluated, to find the optimal model for use in the system.

Dataset availability/freshness will be the biggest risk. This is mitigated by up-to-date uploads.

Stack

- **Frontend:** HTML / JavaScript / Tailwind CSS
- **Backend:** Python / Flask (REST API) – Flask integrates with Python ML models
- **Database:** SQLite, lightweight and no need for db server.
- **ML/Libraries:** scikit-learn (regression models, e.g. Random Forest, Multiple Linear Regression, Gradient Boosting), pandas, joblib, matplotlib (visualisation). Tree based models work well with feature rich datasets.
- **Hosting:** Localhost, (optional cloud: Render)

5. Data & Resources Required

Dataset: Food Demand Forecasting dataset, containing 77 centres, 51 meals, and 145 weeks of orders, and includes prices and promotions. The dataset originates from a Machine Learning hackathon held in 2018, run by Analytics Vidhya and Genpact [3], available on Kaggle under a GPL-2.0 licence [4], and free to use for educational purposes.

Labelled Data: Dataset is structured with target variable (weekly orders). Standard pre-processing and feature engineering will be used in line with the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework.

Compute: No GPU or paid API service required. Local computer or cloud deployment is sufficient.

6. Evaluation

Testing: Unit testing will be performed using pytest, within the Python backend, to test feature engineered variables, such as ‘previous weeks orders’ and ‘discount’, and check there is no data leakage in these features.

Model evaluation: Models will be evaluated using a time based train/validation/test split, using regression metrics like Root Mean Squared Logarithmic Error (RMSLE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2).

End user: A walkthrough of the frontend will be carried out, acting as a manager role to test UX.

Target Success Criteria:

- Evaluation scoring: The chosen model should achieve an RMSLE below 0.55, and an R^2 above 0.80, and should outperform the ‘previous week’ baseline.
- Frontend: The Manager should be able to retrieve the prediction of the following week’s meal orders within three clicks.
- Performance: Results should be displayed within 2 seconds

7. Project Plan & Timeline

The MVP forecasting recommendation system will be completed by week 8, with further weeks to develop the database update feature and refinement.

Phase (Weeks)	Key Deliverables
Wk 1–3	Requirements review, dataset sourcing, SQLite schema, and proof-of-concept exploration.
Wk 4–8	Core build. Data preprocessing, feature engineering, model training, comparison and evaluation. Flask API, SQLite data seeding, Frontend UI build and model integration. MVP completed by week 8.
Wk 8–11	Adjust-plan and record actual sales to database. System performance testing and model refinement.
Wk 11–12	Unit and system testing, final report document, presentation, and demo.

References

[1] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[2] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," International Journal of Forecasting, vol. 38, no. 4, pp. 1346–1364, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0169207021001874

[3] Analytics Vidhya and Genpact, "Genpact Machine Learning Hackathon," Analytics Vidhya, 2018. [Online]. Available: https://www.analyticsvidhya.com/datahack/contest/genpact-machine-learning-hackathon-1/

[4] "Food Demand Forecasting," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/ghoshsaptarshi/av-genpact-hack-dec2018